1. Develop a program to blink 5 LEDs back and forth

```
const int ledPins[] = {2, 3, 4, 5, 6}; // Pins for LEDs
const int numLeds = 5;
void setup()
{
  for (int i = 0; i < numLeds; i++)
{
    pinMode(ledPins[i], OUTPUT);
  }
}
void loop()
{
  // Left to right
  for (int i = 0; i < numLeds; i++)
{
    digitalWrite(ledPins[i], HIGH);
    delay(500);
    digitalWrite(ledPins[i], LOW);
  }
  // Right to left
  for (int i = numLeds - 1; i >= 0; i--)
{
    digitalWrite(ledPins[i], HIGH);
    delay(500);
    digitalWrite(ledPins[i], LOW);
  }
}
```

2. Develop a program to interface a relay with Arduino board

```
const int relayPin = 7;
void setup()
{
  pinMode(relayPin, OUTPUT);
  digitalWrite(relayPin, LOW);
}

void loop()
 {
  digitalWrite(relayPin, HIGH);
```

```
  delay(2000);
  digitalWrite(relayPin, LOW);
  delay(2000);
}
```

3. Develop a program to deploy an intrusion detection system using Ultrasonic and sound sensors

```
const int trigPin = 6;
const int echoPin = 7;
const int soundPin = 10;
const int buzzer = 4;

long duration;
float distance;

void setup()
{
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
  pinMode(soundPin, INPUT);
  pinMode(buzzer, OUTPUT);
  Serial.begin(9600);
}
void loop()
 {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2;
  int soundVal = digitalRead(soundPin);
  Serial.print("Distance: ");
  Serial.println(distance);
  Serial.print("sound: ");
  Serial.println(soundVal);
  if ((distance > 0 && distance < 20) || soundVal == LOW)
  {
    digitalWrite(buzzer, HIGH);
```

```
    Serial.println("Intrusion Detected!");
    delay(400);
  } else {
    digitalWrite(buzzer, LOW);
    delay(200);
  }
}
```

4. Develop a program to control a DC motor with Arduino board

```
const int in1 = 8;
 const int in2 = 9;
  void setup()
{
    pinMode(in1, OUTPUT);
    pinMode(in2, OUTPUT);
  }
  void loop()
{
    digitalWrite(in1, HIGH);
    digitalWrite(in2, LOW);
    delay(3000);
    digitalWrite(in1, LOW);
    digitalWrite(in2, HIGH);
    delay(3000);
  }
```

5. Develop a program to deploy smart street light system using LDR sensor

```
const int ldrDigitalPin = 2;
const int lightPin = 9;
void setup()
{
  pinMode(ldrDigitalPin, INPUT);
  pinMode(lightPin, OUTPUT);
  Serial.begin(9600);
}
```

```
void loop()
{
 int ldrState = digitalRead(ldrDigitalPin);
 Serial.print("LDR State: ");
 Serial.println(ldrState);

 if (ldrState == HIGH)
{
   digitalWrite(lightPin, HIGH);
 } else
{
   digitalWrite(lightPin, LOW);
 }

 delay(500);
}
```

6. Develop a program to classify dry and wet waste with the Moisture sensor (DHT22)

```
const int moisturePin = A0;
          int moistureValue = 0;
         const int threshold = 400;
         void setup()
         {
         Serial.begin(9600);
         pinMode(moisturePin, INPUT);
         }
         void loop()
         {
          moistureValue = analogRead(moisturePin);
          Serial.print("Moisture Value: ");
          Serial.println(moistureValue);
        if (moistureValue > threshold)
        {
        Serial.println("Dry Waste Detected");
        }
        else
       {
```

```
        Serial.println("Wet Waste Detected");
        }
        delay(1000);
        }
```

7. Develop a program to read the pH value of a various substances like milk, lime and water

```
int sensorPin = A0;
const int samples = 10;
int readings[samples];
int temp;
unsigned long avgValue;

const float V_7 = 4.10;
const float slope = -5.0;
const float calibration_value = 7 - slope * V_7;
void setup()
{
 Serial.begin(9600);
}

void loop() {

 for (int i = 0; i < samples; i++)
 {
  readings[i] = analogRead(sensorPin);
  delay(30);
 }

 for (int i = 0; i < samples - 1; i++)
 {
  for (int j = i + 1; j < samples; j++)
  {
   if (readings[i] > readings[j]) {
    temp = readings[i];
    readings[i] = readings[j];
    readings[j] = temp;
   }
  }
 }

 avgValue = 0;
```

```
  for (int i = 2; i < samples - 2; i++)
   {
    avgValue += readings[i];
  }
  avgValue /= (samples - 4);

  float voltage = avgValue * 5.0 / 1024.0;

  float pHValue = slope * voltage + calibration_value;
  Serial.print("Analog: ");
  Serial.print(avgValue);
  Serial.print(" | Voltage: ");
  Serial.print(voltage, 3);
  Serial.print(" V | pH: ");
  Serial.println(pHValue, 2);
  delay(1000);
}
```

8. Develop a program to detect the gas leakage in the surrounding environment

```
const int gasSensorPin = A0;
const int buzzerPin = 8;
int gasLevel = 0;
int threshold = 400;
void setup()
{
  pinMode(gasSensorPin, INPUT);
  pinMode(buzzerPin, OUTPUT);
  Serial.begin(9600);
}
void loop()
 {
  gasLevel = analogRead(gasSensorPin);
  Serial.print("Gas Level: ");
  Serial.println(gasLevel);
  if (gasLevel > threshold)
{
    digitalWrite(buzzerPin, HIGH);
    Serial.println("Gas Leak Detected!");
  }
```

```
else
{
   digitalWrite(buzzerPin, LOW);
  }
  delay(1000);
}
```

9. Develop a program to demonstrate weather station readings using Arduino

```
#include "DHT.h"
#define DHTPIN 2
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
void setup()
{
  Serial.begin(9600);
  dht.begin();
  Serial.println("Weather Station Initialized");
}
void loop()
{
  float temp = dht.readTemperature();
  float hum = dht.readHumidity();
  if (isnan(temp) || isnan(hum))
{
    Serial.println("Failed to read from DHT sensor!");
    return;
  }
  Serial.print("Temperature: ");
  Serial.print(temp);
  Serial.println(" °C");
  Serial.print("Humidity: ");
  Serial.print(hum);
  Serial.println(" %");

  Serial.println("----------------------");
  delay(2000);
}
```

10. Develop a program to setup a UART protocol and pass a string through the protocol

```
void setup()
{
  Serial.begin(9600);
  delay(1000);
}

void loop()
{
  String message = "Hello from Arduino via UART!";
  Serial.println(message);
  delay(2000);
}
```

To Receive UART Data:

```
void setup()
{
Serial.begin(9600);
}
void loop()
{
  if (Serial.available())
{
    String received = Serial.readString();
    Serial.print("Received: ");
    Serial.println(received);
  }
}
```

11. Develop a water level depth detection system using Ultrasonic sensor

```
const int trigPin = 9;
const int echoPin = 8;
const int MAX_DISTANCE_CM = 100;
long duration;
```

```
int distance;
int waterLevel;
void setup()
{
  Serial.begin(9600);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
}
void loop()
{
  // Trigger ultrasonic pulse
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = duration * 0.034 / 2;
  if (distance > MAX_DISTANCE_CM)
{
    distance = MAX_DISTANCE_CM;
  }
  waterLevel = MAX_DISTANCE_CM - distance;
  Serial.print("Distance to water: ");
  Serial.print(distance);
  Serial.println(" cm");
  Serial.print("Water Level: ");
  Serial.print(waterLevel);
  Serial.print(" cm ");
  Serial.print(waterLevel * 100);
  Serial.println("%");
  Serial.println("-----------------------");
  delay(2000);
}
```

12. Develop a program to simulate interfacing with the keypad module to record the keystrokes

```
#include <Keypad.h>
const byte ROWS = 4;
const byte COLS = 4;
```

```
char hexaKeys[ROWS][COLS] = {
  {'1','2','3','A'},
  {'4','5','6','B'},
  {'7','8','9','C'},
  {'*','0','#','D'}
};

byte rowPins[ROWS] = {9, 8, 7, 6};//R1=9,R2=8,R3=7,R4=6
byte colPins[COLS] = {5, 4, 3, 2};//C1=5,C2=4,C3=3,C4=2

Keypad customKeypad = Keypad(makeKeymap(hexaKeys), rowPins, colPins, ROWS,
COLS);
void setup()
{
  Serial.begin(9600);
  Serial.println("Hex Pad Ready");
}

void loop()
{
  char customKey = customKeypad.getKey();

  if (customKey)
{
    Serial.print("Pressed: ");
    Serial.println(customKey);
  }
}
```